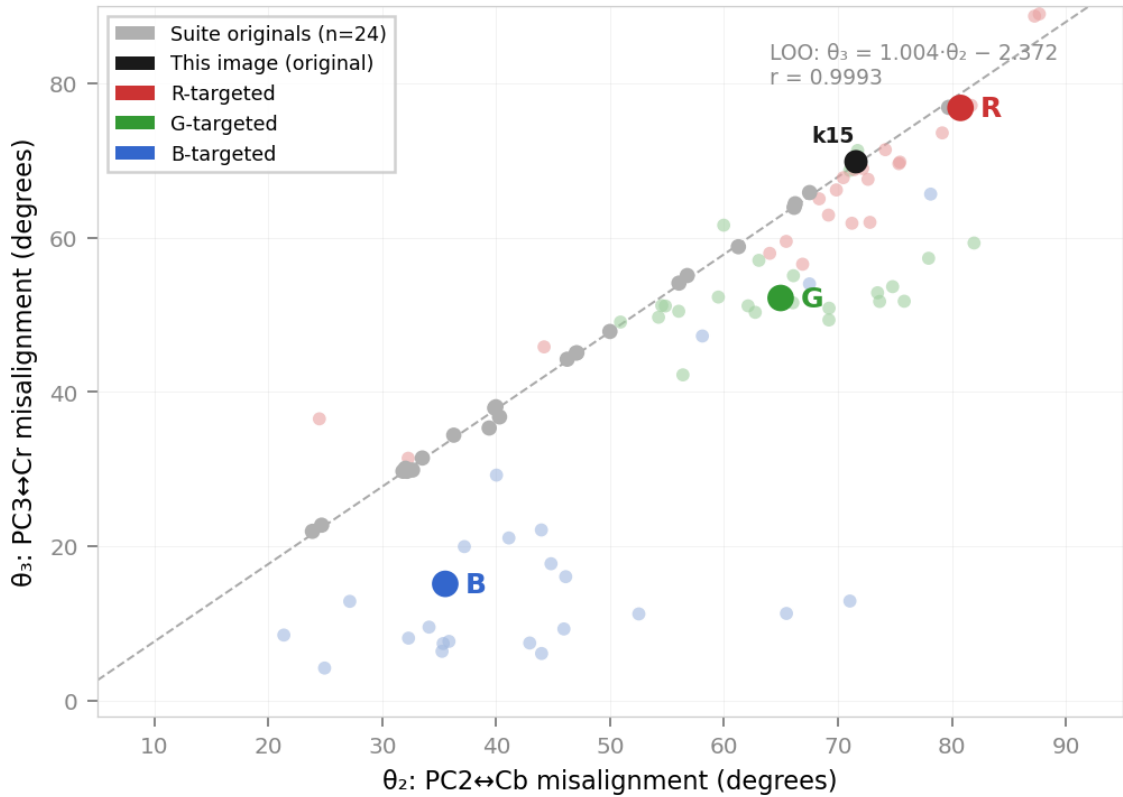


Compression Profile: KODIM15 768x512 | 24-bit RGB | PNG (lossless)

girl with painted face | Strongly One-Dimensional (PC1 = 94.46%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.8602	0.8562	0.9880	0.901	71.58°	69.87°	+0.38°
R-targeted	-0.2405	-0.1205	0.9703	0.444	80.76°	76.85°	-1.86°
G-targeted	0.1219	0.9729	0.1493	0.415	64.97°	52.2°	-10.66°
B-targeted	0.9381	0.2284	0.0874	0.418	35.52°	15.13°	-18.16°

R-targeted isolates red (R-G, R-B < 0.24) while preserving G-B coupling at 0.970.

G-targeted isolates green (R-G, G-B < 0.15) while preserving R-B at 0.973.

B-targeted preserves R-G at 0.938 while driving R-B and G-B near zero.

All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM15768x512 | 24-bit RGB | PNG (lossless)

girl with painted face | Strongly One-Dimensional (PC1 = 94.46%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

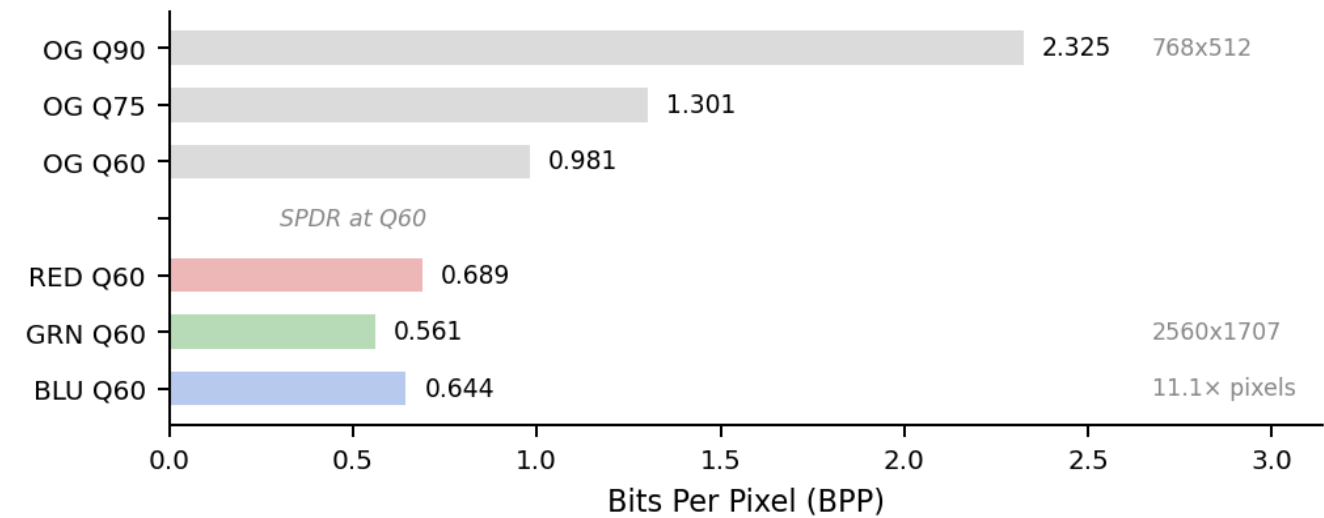
ORIGINAL (kodim15)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	598.2 KB	12.463	393,216
JPEG Q90	JPEG	768x512	111.6 KB	2.325	393,216
JPEG Q75	JPEG	768x512	62.4 KB	1.301	393,216
JPEG Q60	JPEG	768x512	47.1 KB	0.981	393,216

REDISTRIBUTED (kodim15 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	367.3 KB	0.689	4,369,920
GRN Q60	JPEG	2560x1707	299.2 KB	0.561	4,369,920
BLU Q60	JPEG	2560x1707	343.4 KB	0.644	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1× the original pixel count.

Compression Profile: KODIM15 768x512 | 24-bit RGB | PNG (lossless)

girl with painted face | Strongly One-Dimensional (PC1 = 94.46%)

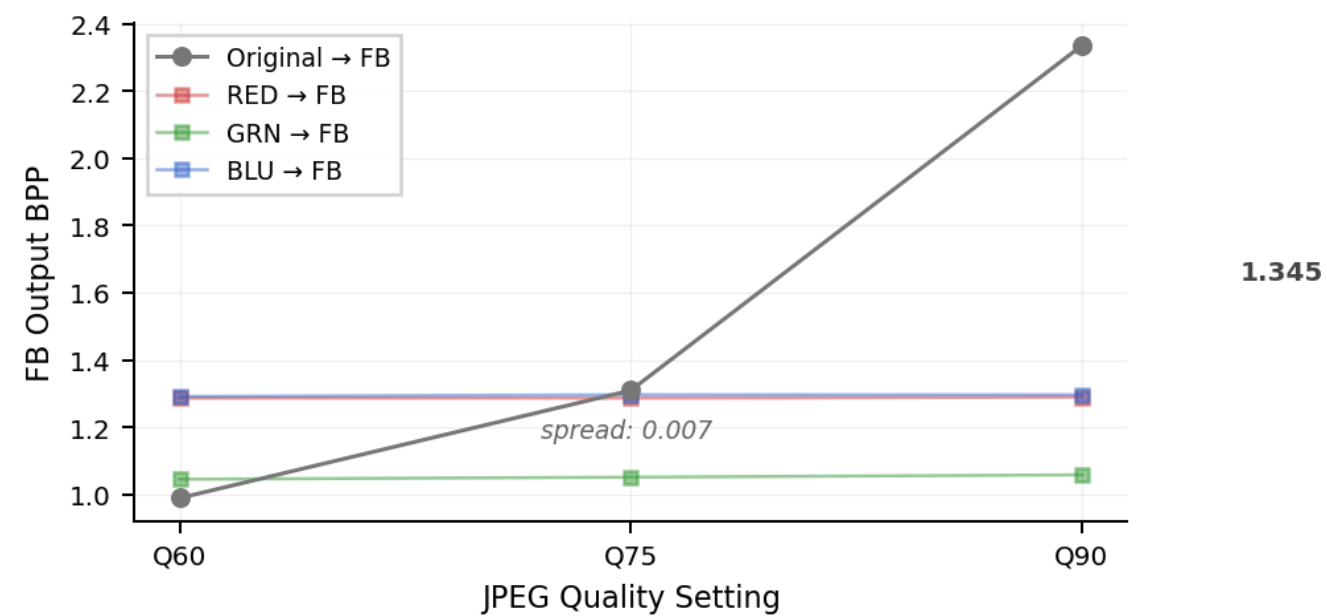
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.124	–
R-targeted → FB	2560x1707	2048x1366	1.290	39.3%
G-targeted → FB	2560x1707	2048x1366	1.059	50.1%
B-targeted → FB	2560x1707	2048x1366	1.297	38.9%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.166	45.1%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9014 vs 0.901 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.345 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.007 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim15

Best BPP (Q60, direct)	0.561	2560x1707, 4,369,920 px
Best BPP through FB	1.059	2048x1366 output
vs OG Q90 → FB	54.6%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216